

Bharatiya Antariksh Hackathon 2026

VyomOS - Autonomous Satellite Intelligence System

Team Name: _____

Team Leader Name: _____

Problem Statement: Antigravity Track (Autonomous Satellite Intelligence System)

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Team Members

Team Leader:

Name: _____ College: _____

Team Member-1:

Name: _____ College: _____

Team Member-2:

Name: _____ College: _____

Team Member-3:

Name: _____ College: _____

Opportunity

How different is it from any of the other existing ideas?

Traditional operations rely on humans reading 2D telemetry dashboards. VyomOS shifts this paradigm by moving intelligence to the edge, processing anomalies autonomously using a local LLM, and providing mitigating commands instantly without internet dependency.

How will it be able to solve the problem?

It reduces cognitive load and reaction time from minutes to milliseconds. By automating detection, explanation, and mitigation, VyomOS prevents catastrophic failures during events like solar flares.

USP of the proposed solution:

- 100% Data Sovereignty (Zero reliance on OpenAI/Clouds)
- Multi-Satellite Cascading Anomaly Correlation
- Live integration with NASA DONKI and real Celestrak TLEs
- Predictive ML forecasting for anomaly severity

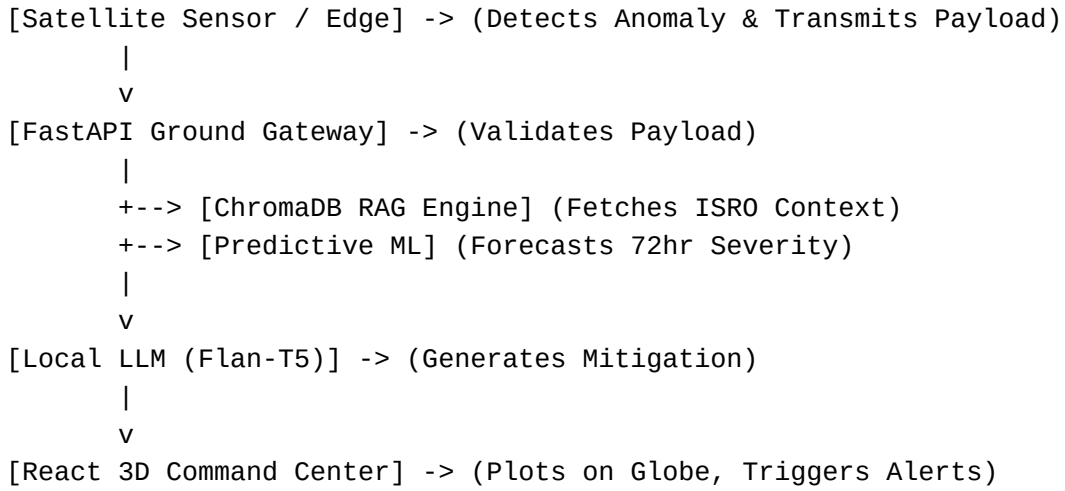
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List of features offered by the solution

- 3D Digital Twin: Real-time WebGL visualization of ISRO's fleet using live TLEs.
- Autonomous AI Mitigation: On-device LLM (Flan-T5) generates instant recovery strategies.
- ISRO RAG Engine: Mitigation is grounded in actual ISRO mission protocols using ChromaDB.
- Predictive Forecasting: ML model predicts anomaly severity 72 hours into the future.
- NASA DONKI Integration: Automatically triggers alerts based on real space weather.
- Telemetry Gauges & Heatmaps: Live tracking of battery, solar output, and historical anomaly density.
- Voice Alert System: Auditory warnings for critical events.
- Automated Incident Reports: One-click PDF generation for mission control logging.

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Process flow diagram / Use-case diagram

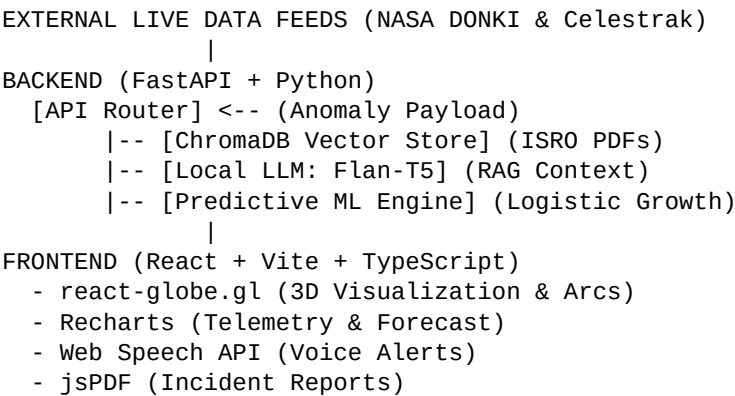


Wireframes / Mock diagrams of the proposed solution

Command Center Layout:

1. Top Bar: Ticker tape showing live X-Ray Flux and Flare Classes, and an Alert Banner.
2. Left Panel: Satellite status matrix, NavIC impact, and live Telemetry Gauges (Recharts).
3. Center/Right: A massive interactive 3D Globe rendering ISRO satellites, orbital paths, Day/Night terminators, and pulsing Red Alert shockwaves.
4. Bottom Panel: 72-hour predictive anomaly severity chart.
5. Floating Assistant: A chat window powered by Gemini 3.5 Flash for ad-hoc space weather queries.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Frontend:

- React 18, TypeScript, Vite
- Tailwind CSS (Glassmorphic UI)
- react-globe.gl & three.js (3D rendering)
- recharts & d3 (Data visualization)
- satellite.js (Orbital mechanics)

Backend:

- Python 3.10+, FastAPI, Uvicorn
- Pydantic (Data validation)

AI & Machine Learning:

- chromadb (Vector Database for RAG)
- sentence-transformers (Embeddings)
- Google flan-t5-small (On-device LLM)
- Gemini 3.5 Flash (Advisory Generation)
- scipy & numpy (Predictive forecasting)

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Estimated implementation cost

Development & Software Cost:

- \$0 (VyomOS uses open-source frameworks, free public APIs like NASA DONKI, and local AI models).

Deployment Hosting (Command Center):

- \$0 for Hackathon (Render/Vercel Free Tiers).
- ~\$50/month (Production-grade cloud backend).

Edge Hardware (Per Satellite/Node):

- ~\$150 (NVIDIA Jetson Nano or equivalent space-hardened edge compute module for running the local LLM on-orbit).

Total Prototype Cost: \$0